

# Project Idea: SonicBot

For Hackclub

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I want to build a small standalone sample keyboard for under \$100.

Instead of making pads, it will have 25 physical keys about two octaves. I can load a WAV sample onto a microSD card then play that sample across the keyboard at different pitches. I could load a vocal phrase, a robotic sound, a bass or a synth. Each key plays the same sound at a different note, turning one sample into a playable instrument.

## What It Does

- Plays WAV samples from a microSD card
- Has 25 handmade keyboard keys
- Changes sample pitch across the keys
- Has knobs for:
  - Attack
  - Release
  - Filter
  - Glide/slide
- Has octave up/down buttons
- Outputs sound through headphones or speakers
- Connects to a computer through USB MIDI

The first version will be monophonic, meaning one note at a time. That is perfect for basslines, melodic leads, robotic vocals, and sound effects. Later, it can be upgraded to play chords

## Hardware

| Part                     | Purpose                                                | Estimated Cost |
|--------------------------|--------------------------------------------------------|----------------|
| Teensy 4.0               | Main brain: reads keys, controls MIDI, processes audio | \$25-30        |
| Teensy Audio Shield      | High-quality audio output and microSD support          | \$10-18        |
| MicroSD card             | Stores WAV samples                                     | \$5-8          |
| 25 handmade keys         | The keyboard itself                                    | \$10-20        |
| 25 diodes                | Prevents false notes when multiple keys are pressed    | \$2-4          |
| 4 potentiometers         | Attack, Release, Filter, Glide                         | \$4-7          |
| Buttons, wire, perfboard | Octave controls and wiring                             | \$8-12         |
| DIY enclosure            | Wood, acrylic, 3D print, or recycled plastic           | \$0-10         |

Target total: around \$80-100.

Teensy's audio tools are designed for sampled playback, synthesis, effects, filtering, recording, and musical audio projects, which makes it a good fit for a compact standalone instrument. Its Audio Shield adds stereo audio, headphone output, microSD storage, and 16-bit/44.1 kHz sound capability

## Building the Keys From Scratch

The most original part of this project is making the keyboard keys myself

Rather than buying piano keys, I'll build 25 individual lever keys from wood, acrylic, or 3D-printed plastic, so when I press the key:

1. The key pivots on a small axle, screw, rod, or hinge.
2. The back of the key moves downward.
3. It presses a button
4. The Teensy detects the key press and plays the assigned note.
5. A spring, foam, rubber band, or flexible plastic returns the key to its resting position.

## Simple materials for the keys

- Thin plywood, acrylic, or 3D-printed strips for white keys
- Smaller black pieces placed higher for sharps flats
- Wooden dowels, skewers, screws, or metal rods as pivots
- Foam, silicone, springs, or rubber bands for key return
- Hot glue, screws, or epoxy for assembly

## Keyboard Layout

Visually, I would build it like a real keyboard:

Black keys:   
 White keys: C D E F G A B C

The white keys can be longer and lower. The black keys can be shorter, raised, and mounted behind them. This will make the instrument look like a real mini synthesizer rather than a row of buttons.

## Electronics Design

Every key needs a switch and diode. The 25 keys are wired as a **5 × 5 matrix**, so the Teensy only uses 10 input/output pins instead of 25.

|        | Columns |     |     |     |     |
|--------|---------|-----|-----|-----|-----|
|        | 1       | 2   | 3   | 4   | 5   |
| Rows 1 | Key     | Key | Key | Key | Key |
| Rows 2 | Key     | Key | Key | Key | Key |
| Rows 3 | Key     | Key | Key | Key | Key |
| Rows 4 | Key     | Key | Key | Key | Key |
| Rows 5 | Key     | Key | Key | Key | Key |

Each key includes a diode so multiple notes can be detected correctly without unwanted “ghost” notes. Mechanical keyboard matrices use the same basic approach for reliable input

For the first version, every key can use fixed velocity. That keeps the handmade key mechanism much easier. Later, I can add velocity sensing with two switches per key, Hall sensors, or force sensors.

## Controls and Sound

The front panel will be simple:

[Attack] [Release] [Filter] [Glide]  
[Oct -] [Sample] [Oct +]

- **Attack:** Controls how quickly the sound starts.
- **Release:** Controls how long it fades after releasing a key.
- **Filter:** Controls brightness or darkness of the sample.
- **Glide:** Slides smoothly between notes, useful for bass and lead sounds.
- **Octave buttons:** Shift the 25 keys up or down.
- **Sample button:** Cycles through files on the microSD card.

I'll initially use a few fixed WAV files attached:

01\_VOICE.WAV  
02\_ROBOT.WAV  
03\_BASS.WAV  
04\_TEXTURE.WAV

The key assignment changes the playback rate of the sample. Lower keys play it slower and deeper; higher keys play it faster and brighter.

## First Prototype Goal

The first finished version only needs to do this:

Load one WAV file, play it with 25 handmade keys, and shape it with Attack, Release, Filter, and Glide.

That is enough to create a real, expressive instrument: part sampler, part synthesizer, part robotic cyberpunk thingy art piece. The design can later grow into a polyphonic keyboard or robotic performance instrument.